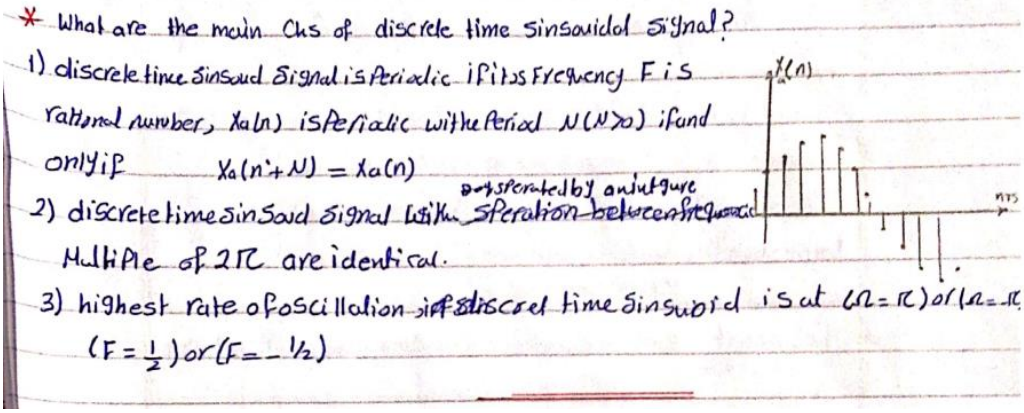
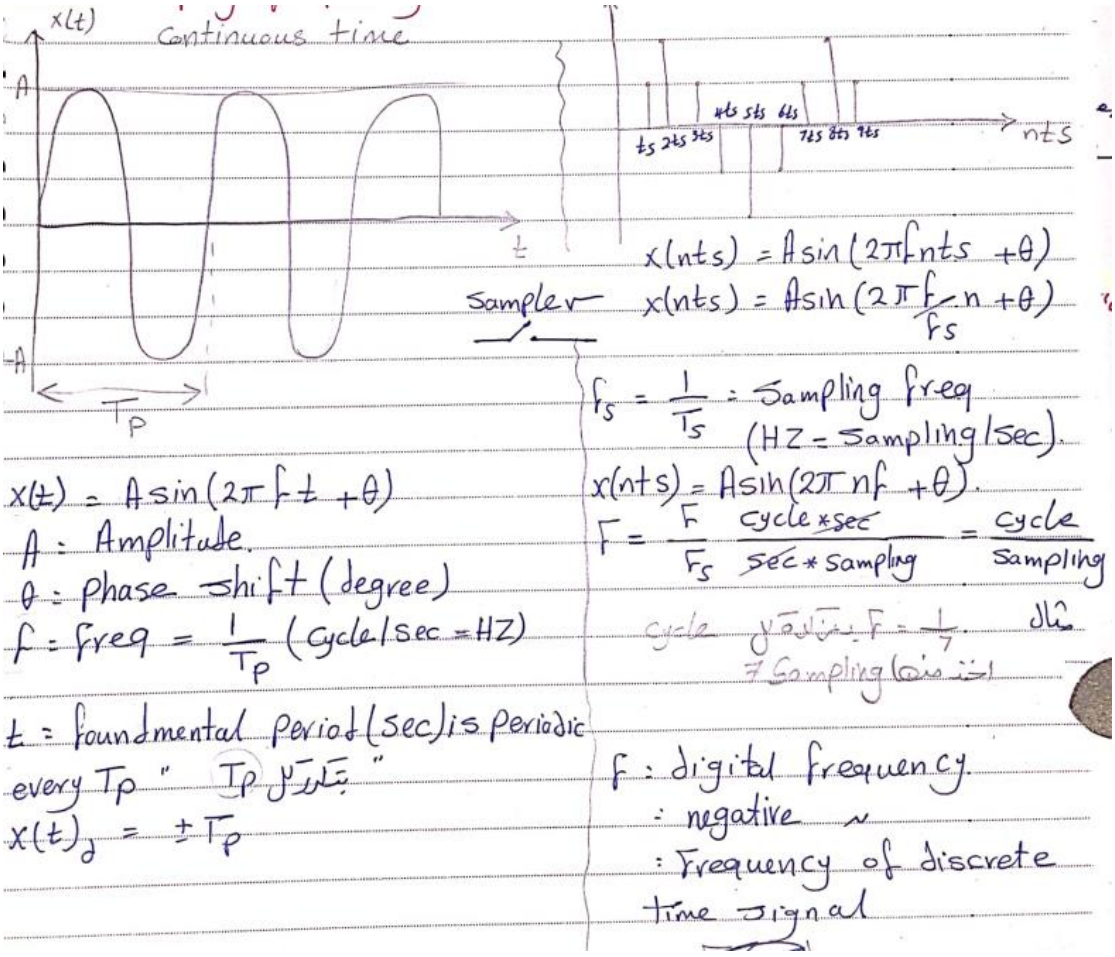
- In detail, what are the main characteristics of the discrete time sinusoidal signals?



- Find the relationship between the frequency of analog signal, frequency of discrete-time signal and sampling frequency.



- Define (using equations and/or figures) the following items: Correlation operation, Correlation coefficient.

II- Define the items : i) Correlation operation: process which show 2 data sequence we can use Correlation to compare to similarity

iii) Correlation Coefficient : used to measure cross correlation according to the fixed scale between -1 & +1.

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For LTIS define (using equations and/or figures) the linearity, stability, ROC and difference equation

(ii) stability :

An LTIS is stable if its impulse response h(k), satisfied the following condition : $\sum_{k=-\infty}^{\infty} h(k) < \infty$

This condition is satisfied if, h(k) is of finite duration. or if h(k) delays towards zero as k increase.

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iv) RoC : Region of convergence

it's the region where the Z-transform converges & in this region the values of x(z) are finite

BIBO : "bounded i/p bounded o/p"

An LTI system is said to be BIBO stable if and only if it satisfies criterion

$$\sum_{k=0}^{\infty} |h(k)| < \infty$$

h(k) : impulse with response the system

Time invariant system :

A discrete time signal is said to be time invariat if it is output is independent of the time the i/p is applied

$x(n) \longrightarrow y(n)$

$x(n-k) \longrightarrow y(n-k)$

b- Check (β) for the right statements and (X) for the wrong statements and correct the mistake

. i- In quantization process, when the quantization step (Δ) increases, the accuracy of the quantizer increases. (X)

ii- Nyquist frequency (FN), is the minimum sampling frequency (r)

iii- Fourier transform (FT) is calculated for a non-periodic continuous – time signals. (r)

iv- Discrete Fourier transform (DFT) is obtained for the analog signals. (X)

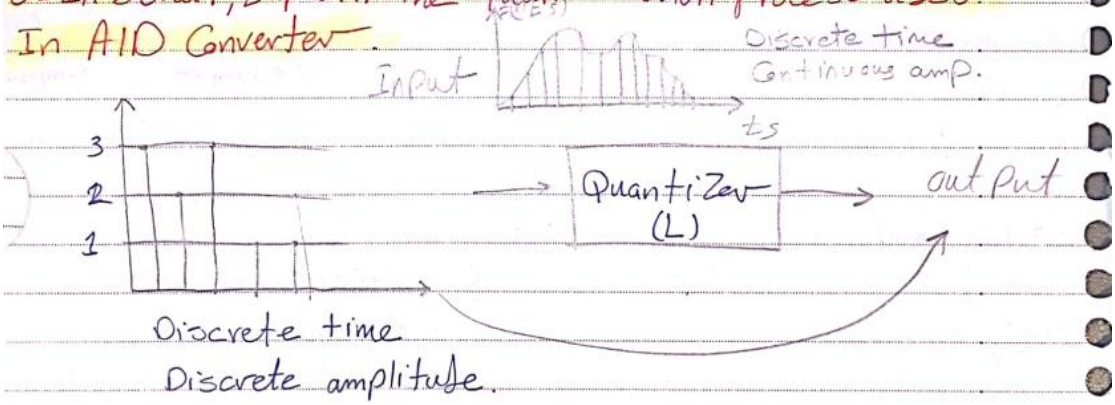
v- To reconstruct the continuous time signal, x (t), from its discrete time sequence, X(n), the sampling rate, FS, must be less than the frequency of x(t) signal, f0. (X)

For LTIS define (using equations and/or figures) the following item: causality and different equation ?

ii) Consality:
The system is used to be causal system if it produces an o/p only where there is an input. all physical systems are causal. In general a causal D.T system has zero impulse response before time zero ($h(k)=0$ for all $k<0$) also the D.T sequence say $x(n)$ is causal if its value is zero befor time zero ($x(n)=0$ for all $n<0$)

→ Difference equation:
specifies the actual operation that must be performed by the discrete time system on the i/p data in the time domain in order to generate the desired o/p.

8- In detail, Explain the quantization process used In A/D Converter.



Quantizer → Convert Continuous amplitude to discrete amplitude by representing each sample value to the nearest level.

Difference between unquantized & quantized signal

$$eq(n) = x_a(n) - x(n) \quad eq(n)$$

$$\Delta = \frac{x_{max} - x_{min}}{L-1}$$

10- prove that the DFT, $x(k)$, of a sequence $x(n)$ is periodic.

$x(n) = x(n+np)$ ← periodic

$x(nts) = x(n) = A\sin(2\pi f n + \theta)$

$x(n+np) = A\sin(2\pi f(n+np) + \theta)$

$= A\sin(2\pi f n + 2\pi f np + \theta)$

$F = \frac{k}{np}$ < if $np = k \rightarrow k$ = integer number.

$\therefore x(n+np) = A\sin(2\pi f n + 2\pi k + \theta)$

$\therefore x(n+np) = A\sin(2\pi f n + \theta)$

$x(n) = x(n+np)$

$\therefore$ Signal is periodic.

→ filter design steps :

1) specification of the filter requirements it include.
   i) signal charactristics (type of signal source and sink, I/o interface datarates & width & highest frequency of interest).
   ii) the charactristics of filter (Amplitude, phase response, tolerance, speed)
   iii) the manner of implementation.
   iv) other design constrains (For ex: the cost of filter).

2) Calculation of suitable filter coefficients.
   - they are several methods of calculating filter coefficients the most widely used
   - Impulse invariant (IIR) . Window (FIR).
   - Bilinear transformation (IIR) . Frequency Sampling (FIR)
   - Pole-Zero placement (IIR) . optimal (FIR).

3) representation of filter by suitable structure (realization)

FIR	IIR
direct	direct form
freq sampling	Cascade
fast Convolution	Parallel
Lattice	

4) Analysis of infite word length effects.
main source of performance degradation in digital filters are
① input/output signal quantization
② Coefficient quantization ③ arithmatic round errors
④ over flow

5) implemetation of a filter.

iii) Correlation Coefficient : used to measure cross correlation according to the fixed scale between -1 & +1.

$$r_{12}(j) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n)\, x_2(n+j)$$

where x1 reference signal
x2 delayed output from circuit

j : response the amount of ptg
N : no of samples

vi) Convolution operation : How the i/p interact with the sys to produce the o/p.

(ii) Correlation theorem : can be used to obtain the linear correlation by adding augmenting zeros to the 2 sequences.